

C-style Strings

- Sequence of characters
 - contiguous in memory
 - implemented as an array of characters
 - terminated by a null character (null)
 - null – character with a value of zero
 - Referred to as zero or null terminated strings
- String literal
 - sequence of characters in double quotes, e.g. "Frank"
 - constant
 - terminated with null character

C-style Strings

declaring variables

```
char my_name[] {"Frank"};
```

F	r	a	n	k	\0
---	---	---	---	---	----

```
my_name[5] = 'y'; // Problem
```

C-style Strings

declaring variables

```
char my_name[8] {"Frank"};
```

F	r	a	n	k	\0	\0	\0
---	---	---	---	---	----	----	----

```
my_name[5] = 'y'; // OK
```

#include <cstring>

Functions that work with C-style Strings

- Copying
- Concatenation
- Comparison
- Searching
- and others

#include <cstring>

A few examples

```
char str[80];  
  
strcpy(str, "Hello "); // copy  
  
strcat(str, "there"); // concatenate  
  
cout << strlen(str); // 11  
  
strcmp(str, "Another"); // > 0
```

Character Functions

Testing characters

we can also use string
instead of C

<code>isalpha(c)</code>	True if c is a letter
<code>isalnum(c)</code>	True if c is a letter or digit
<code>isdigit(c)</code>	True if c is a digit
<code>islower(c)</code>	True if c is lowercase letter
<code>isprint(c)</code>	True if c is a printable character
<code>ispunct(c)</code>	True if c is a punctuation character
<code>isupper(c)</code>	True if c is an uppercase letter
<code>isspace(c)</code>	True if c is whitespace

Character Functions

Converting characters

<code>tolower(c)</code>	returns lowercase of c
<code>toupper(c)</code>	returns uppercase of c